

1

COMP

Membrane composition and fluidity

You should be able to: Describe the fluid mosaic membrane and state how phospholipids, cholesterol, glycolipids, and integral and peripheral proteins contribute to its properties.

PROMPT ☐ A ☐ B PASS 1 COLOUR _____ PASS 2 COLOUR _____

Every one of these boxes is **drawn**. If you need words, they go inside little boxes with arrows between them, in the order things happen. Sentences running across the page do not count.

1 · DRAW IT*a picture, not a paragraph***2 · LABEL EVERY PART***lines out to the drawing***3 · WHAT HAPPENS, IN ORDER***boxes and arrows, step to step***4 · WHAT BREAKS IF THIS FAILS***follow the arrows to what stops*

2

COMP

Determinants of permeability

You should be able to: Rank molecules by their ability to cross a lipid bilayer unaided using size, charge, and lipid solubility, and predict which will require a protein.

PROMPT ☐ A ☐ B PASS 1 COLOUR _____ PASS 2 COLOUR _____

Every one of these boxes is **drawn**. If you need words, they go inside little boxes with arrows between them, in the order things happen. Sentences running across the page do not count.

1 · DRAW IT*a picture, not a paragraph***2 · LABEL EVERY PART***lines out to the drawing***3 · WHAT HAPPENS, IN ORDER***boxes and arrows, step to step***4 · WHAT BREAKS IF THIS FAILS***follow the arrows to what stops*

3

COMP

Simple diffusion and Fick's law

You should be able to: State the variables in Fick's law of diffusion and predict how a change in concentration gradient, surface area, membrane thickness, or distance changes the rate of transfer.

PROMPT ☐ A ☐ B PASS 1 COLOUR _____ PASS 2 COLOUR _____ ALSO A LAB SKILL

Every one of these boxes is **drawn**. If you need words, they go inside little boxes with arrows between them, in the order things happen. Sentences running across the page do not count.

1 · DRAW IT*a picture, not a paragraph***2 · LABEL EVERY PART***lines out to the drawing***3 · WHAT HAPPENS, IN ORDER***boxes and arrows, step to step***4 · WHAT BREAKS IF THIS FAILS***follow the arrows to what stops*

4

COMP

Osmolarity and tonicity

You should be able to: Calculate the osmolarity of a solution, distinguish osmolarity from tonicity, and classify a solution as isotonic, hypotonic, or hypertonic to a cell.

PROMPT ☐ A ☐ B PASS 1 COLOUR _____ PASS 2 COLOUR _____ ALSO A LAB SKILL

Every one of these boxes is **drawn**. If you need words, they go inside little boxes with arrows between them, in the order things happen. Sentences running across the page do not count.

1 · DRAW IT*a picture, not a paragraph***2 · LABEL EVERY PART***lines out to the drawing***3 · WHAT HAPPENS, IN ORDER***boxes and arrows, step to step***4 · WHAT BREAKS IF THIS FAILS***follow the arrows to what stops*

5

COMP

Osmosis and cell volume

You should be able to: Predict the direction of water movement and the resulting change in cell volume when a cell is placed in a solution of stated osmolarity and penetrating solute content.

PROMPT ☐ A ☐ B PASS 1 COLOUR _____ PASS 2 COLOUR _____ ALSO A LAB SKILL

Every one of these boxes is **drawn**. If you need words, they go inside little boxes with arrows between them, in the order things happen. Sentences running across the page do not count.

1 · DRAW IT*a picture, not a paragraph***2 · LABEL EVERY PART***lines out to the drawing***3 · WHAT HAPPENS, IN ORDER***boxes and arrows, step to step***4 · WHAT BREAKS IF THIS FAILS***follow the arrows to what stops*

6

COMP

Diffusion and osmosis experiment

You should be able to: Measure diffusion and osmotic movement across a selectively permeable membrane and relate the observed rate to molecular size and concentration gradient.

PROMPT ☐ A ☐ B PASS 1 COLOUR _____ PASS 2 COLOUR _____ ALSO A LAB SKILL

Every one of these boxes is **drawn**. If you need words, they go inside little boxes with arrows between them, in the order things happen. Sentences running across the page do not count.

1 · DRAW IT*a picture, not a paragraph***2 · LABEL EVERY PART***lines out to the drawing***3 · WHAT HAPPENS, IN ORDER***boxes and arrows, step to step***4 · WHAT BREAKS IF THIS FAILS***follow the arrows to what stops*

7

COMP

Tonicity and red blood cells

You should be able to: Observe erythrocytes in solutions of different tonicity, identify crenation, normal shape, and hemolysis, and explain each result.

PROMPT ☐ A ☐ B PASS 1 COLOUR _____ PASS 2 COLOUR _____ ALSO A LAB SKILL

Every one of these boxes is **drawn**. If you need words, they go inside little boxes with arrows between them, in the order things happen. Sentences running across the page do not count.

1 · DRAW IT*a picture, not a paragraph***2 · LABEL EVERY PART***lines out to the drawing***3 · WHAT HAPPENS, IN ORDER***boxes and arrows, step to step***4 · WHAT BREAKS IF THIS FAILS***follow the arrows to what stops*

8

Facilitated diffusion

COMP

You should be able to: Describe carrier and channel mediated diffusion and explain why carrier mediated transport shows saturation and specificity while simple diffusion does not.

PROMPT ☐ A ☐ B PASS 1 COLOUR _____ PASS 2 COLOUR _____

Every one of these boxes is **drawn**. If you need words, they go inside little boxes with arrows between them, in the order things happen. Sentences running across the page do not count.

1 · DRAW IT*a picture, not a paragraph***2 · LABEL EVERY PART***lines out to the drawing***3 · WHAT HAPPENS, IN ORDER***boxes and arrows, step to step***4 · WHAT BREAKS IF THIS FAILS***follow the arrows to what stops*

9

COMP

Primary active transport

You should be able to: Explain how the sodium potassium ATPase uses ATP to move three sodium out and two potassium in, and state the two gradients it maintains.

PROMPT ☐ A ☐ B PASS 1 COLOUR _____ PASS 2 COLOUR _____ ALSO A LAB SKILL

Every one of these boxes is **drawn**. If you need words, they go inside little boxes with arrows between them, in the order things happen. Sentences running across the page do not count.

1 · DRAW IT*a picture, not a paragraph***2 · LABEL EVERY PART***lines out to the drawing***3 · WHAT HAPPENS, IN ORDER***boxes and arrows, step to step***4 · WHAT BREAKS IF THIS FAILS***follow the arrows to what stops*

10

COMP

Secondary active transport

You should be able to: Distinguish symport from antiport and trace how the sodium gradient powers glucose uptake by SGLT and calcium removal by the sodium calcium exchanger.

PROMPT ☐ A ☐ B PASS 1 COLOUR _____ PASS 2 COLOUR _____

Every one of these boxes is **drawn**. If you need words, they go inside little boxes with arrows between them, in the order things happen. Sentences running across the page do not count.

1 · DRAW IT*a picture, not a paragraph***2 · LABEL EVERY PART***lines out to the drawing***3 · WHAT HAPPENS, IN ORDER***boxes and arrows, step to step***4 · WHAT BREAKS IF THIS FAILS***follow the arrows to what stops*

11

COMP

Transport maximum and saturation

You should be able to: Interpret a transport rate curve, identify the transport maximum, and apply the concept to renal glucose handling in hyperglycemia.

PROMPT ☐ A ☐ B PASS 1 COLOUR _____ PASS 2 COLOUR _____ ALSO A LAB SKILL

Every one of these boxes is **drawn**. If you need words, they go inside little boxes with arrows between them, in the order things happen. Sentences running across the page do not count.

1 · DRAW IT*a picture, not a paragraph***2 · LABEL EVERY PART***lines out to the drawing***3 · WHAT HAPPENS, IN ORDER***boxes and arrows, step to step***4 · WHAT BREAKS IF THIS FAILS***follow the arrows to what stops*

12

COMP

Vesicular transport

You should be able to: Compare phagocytosis, pinocytosis, receptor mediated endocytosis, and exocytosis by trigger, cargo, and energy requirement.

PROMPT ☐ A ☐ B PASS 1 COLOUR _____ PASS 2 COLOUR _____

Every one of these boxes is **drawn**. If you need words, they go inside little boxes with arrows between them, in the order things happen. Sentences running across the page do not count.

1 · DRAW IT

a picture, not a paragraph

2 · LABEL EVERY PART

lines out to the drawing

3 · WHAT HAPPENS, IN ORDER

boxes and arrows, step to step

4 · WHAT BREAKS IF THIS FAILS

follow the arrows to what stops

13

COMP

Transepithelial transport

You should be able to: Trace glucose or sodium from lumen to blood across a polarized epithelium and identify which step occurs at the apical membrane and which at the basolateral membrane.

PROMPT ☐ A ☐ B PASS 1 COLOUR _____ PASS 2 COLOUR _____

Every one of these boxes is **drawn**. If you need words, they go inside little boxes with arrows between them, in the order things happen. Sentences running across the page do not count.

1 · DRAW IT*a picture, not a paragraph***2 · LABEL EVERY PART***lines out to the drawing***3 · WHAT HAPPENS, IN ORDER***boxes and arrows, step to step***4 · WHAT BREAKS IF THIS FAILS***follow the arrows to what stops*

14

COMP

Transport simulation

You should be able to: Use a membrane transport simulation to distinguish simple diffusion from facilitated diffusion and active transport by their response to gradient reversal and metabolic poison.

PROMPT ☐ A ☐ B PASS 1 COLOUR _____ PASS 2 COLOUR _____ ALSO A LAB SKILL

Every one of these boxes is **drawn**. If you need words, they go inside little boxes with arrows between them, in the order things happen. Sentences running across the page do not count.

1 · DRAW IT*a picture, not a paragraph***2 · LABEL EVERY PART***lines out to the drawing***3 · WHAT HAPPENS, IN ORDER***boxes and arrows, step to step***4 · WHAT BREAKS IF THIS FAILS***follow the arrows to what stops*